



Computer Vision Assignment IV

Detection and tracking

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Motivation and Context

Object detection and tracking in aerial images (UAVs/Drones) are critical tasks in modern surveillance, traffic management, and search-and-rescue operations. Unlike standard datasets, aerial views present unique challenges: small object scales, high density of targets, and significant camera motion. Traditional tracking methods relied on simple geometric associations. However, modern approaches integrate Deep Learning detectors with sophisticated data association strategies. This project aims to explore the state-of-the-art (SOTA) in Multi-Object Tracking (MOT) by decoupling the detection and association phases.

Dataset Architecture and Protocols

The **VisDrone2019-MOT-val** validation subset will be used. This dataset follows a strict hierarchy that students must respect for a valid evaluation.

Folder Structure

Upon extracting the dataset, the following folders will be found:

- **sequences/**: Contains subfolders for each video (e.g., `uav0000086_00000_v`). Inside are sequentially numbered JPEG images (`0000001.jpg`, etc.). These are the input data for the pipeline.
- **annotations/**: Contains the **Ground Truth (GT)**. Each `.txt` file is a manually constructed label. It is used exclusively for metrics calculation.
- **det/** (Ignore): Contains pre-calculated detections using legacy models. **These must not be used.** The core of the project is to generate new detections using modern models.

MOT Challenge Annotation Format

Both the GT and the students' results must follow the MOT standard:

```
<frame>, <id>, <bb_left>, <bb_top>, <bb_width>, <bb_height>, <conf>, <class>, <vis>
```

1. **frame**: Frame index (starts at 1).
2. **id**: Unique trajectory identifier (must be persistent).
3. **bb_left**, **bb_top**: (x, y) coordinates of the top-left corner.
4. **bb_width**, **bb_height**: Bounding box dimensions in pixels.
5. **conf**: Detector confidence (usually 1 in the GT).
6. **class**: Object category. **Restriction:** Filter results only for **Class 1 (Pedestrian)** and **Class 4 (Car)**.

Methodology

Students must implement a modular pipeline. The use of "all-in-one" functions (such as `model.track()`) is not allowed without a detailed justification of the component integration.

Model Comparison

Four different configurations must be compared:

- **Method 1 (Baseline)**: YOLOv11 + SORT. A purely geometric approach.
- **Method 2 (Efficiency)**: YOLO26 + ByteTrack. Analysis of NMS removal on tracking latency.
- **Method 3 (Accuracy)**: RT-DETR + BoT-SORT. Use of global attention and Global Motion Compensation (GMC).
- **Method 4 (Resolution)**: Any of the above integrating **SAHI (Slicing Aided Hyper Inference)** to detect small objects.

Note that the ideal approach would be to train the models; however, since training with Ultralytics is now very straightforward, the task requires a two-step process: first, detection, and then bounding box matching. While some proposed algorithms in the pipeline do not use visual features, others do; this is where your development comes in—you must tune the tracking algorithm. Furthermore, since the images are from drones, it is not enough to use standard algorithms without strategies addressing small object detection. For this, you may use the most suitable strategy, such as specialized training, searching for pre-trained models online, etc. This will be demonstrated in fourth method.

Evaluation Protocol

Quantitative evaluation will be performed using the `motmetrics` library. The following averaged indicators must be reported:

- **MOTA (Multiple Object Tracking Accuracy):** Accumulated error of false positives, false negatives, and ID switches.
- **IDF1 (ID F1-score):** The model's ability to maintain identity over the long term.
- **ID Switches (IDS):** Total number of times an object changes its identifier.
- **FPS (Frames Per Second):** Temporal performance of the entire system (Inference + Association).

Deliverables

An oral classroom presentation supported by slides, including:

- Technical explanation of the **YOLO26** vs. **RT-DETR** architecture.
- Justification for using appearance (embeddings) versus purely geometry in tracking.
- Analysis of the impact of **SAHI** on low-resolution object detection.
- Visual demos showing success cases and failures (occlusions).

Rubric

For this practice, only the presentation is required; a written document is not necessary. The rubric is as follows:

- (5pts) Presentation: Slide content and formality.
- (7pts) Knowledge: Semantic content of the slides.
- (3pts) Q&A: Questions answered during the presentation.
- (5pts) Innovation: Heuristics used to improve results.